

Jeffrey D. Walker
Yale University

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CURRENT POSITION

Associate Research Scientist, Saxena Lab
Yale University, Department of Biomedical Engineering
Wu Tsai Institute, Center for Neurocomputation and Machine Intelligence

October 2024 – present

EDUCATION

University of Chicago

PhD, Computational Neuroscience

Dissertation: Methods for studying the neural control of natural behaviors using unconstrained marmosets

Advisor: Nicholas Hatsopoulos, Professor Organismal Biology & Anatomy

Chicago, IL
October 2012 - July 2018

Wesleyan University

MA, Neuroscience and Behavior

Thesis: The Propagation of epileptiform events across the corpus callosum

Advisor: Gloster Aaron, Associate Professor Neuroscience and Behavior

Middletown, CT
2008 – 2009

BA, Honors, Neuroscience and Behavior

Thesis: The Interface of phenomenology and the cognitive sciences

Advisor: Steve Horst, Professor of Philosophy

2004 – 2008

RESEARCH EXPERIENCE

University of Chicago, Organismal Biology & Anatomy

Postdoctoral Scholar; Advisors: Nicholas Hatsopoulos and Jason MacLean

Sensorimotor cortical population dynamics across marmoset natural behaviors and sleep

2018 – 2024

University of Chicago, Committee on Computational Neuroscience

Graduate Research Assistant; Advisor: Nicholas Hatsopoulos

Methods to study the neural control of movement with common marmosets

2012 – 2018

Wesleyan University, Biology Department

Student then lab technician; Advisor: Gloster Aaron

Propagation of epileptiform events across the corpus callosum

2007 – 2010

UC Davis Center for Mind and Brain

Intern; Supervisor: Cliff Saron

Summer 2006

PUBLICATIONS

†:co-senior author, *:co-first author

Almani N*, Lazzari J*, **Walker J***, Saxena S (2026) Embodied sensorimotor control: computational modeling of the neural control of movement. *Annual Reviews in Biomedical Engineering*. 10.1146/annurev-bioeng-102723-020454.

Aparicio P*, **Walker J***, MacLean J, Hatsopoulos N (2025) Spatially extensive LFP correlations identify slow-wave sleep in marmoset sensorimotor cortex. *eNeuro*. 10.1523/ENEURO.0139-25.2025

- Moore D, MacLean J, **Walker J**†, Hatsopoulos N† (2024) A dynamic subset of network interactions underlies tuning to natural movements in marmoset sensorimotor cortex. *Cell Reports*. 15:10517
- Moore D, **Walker J**, MacLean J, Hatsopoulos N (2022) Validating markerless pose estimation with 3D X-ray radiography. *Journal of Experimental Biology*. 225, jeb243998. 10.1242/jeb.243998.
- Walker J**, Pirschel F, Sundiang M, Niekrasz M, MacLean J, Hatsopoulos N (2021) Chronic wireless neural population recordings with common marmosets. *Cell Reports* 36(2):109379. 10.1016/j.celrep.2021.109379.
- Walker J**, Pirschel F, Gidmark N, MacLean J, Hatsopoulos N (2020) A platform for semi-automated voluntary training of common marmosets for behavioral neuroscience. *Journal of Neurophysiology* 123(4):1420-1426. 10.1152/jn.00300.2019.
- Walker J**, MacLean J, Hatsopoulos N (2017) The marmoset as a model system for studying voluntary motor control. *Developmental Neurobiology*. 77(3):273-285. 10.1002/dneu.22461
- Walker J**, Storch G, Quach-Wong B, Sonnenfeld J, Aaron G (2012) Propagation of epileptiform events across the corpus callosum in a cortical slice preparation. *PLOS ONE*. 7(2):e31415. 10.1371/journal.pone.0031415
- Aaron G, Briggs S, **Walker J**, Asik K, Lombroso P, Naegel J (2011) STEP Regulation of seizure thresholds in the hippocampus. *Epilepsia* 52(3):497-506.

MANUSCRIPTS IN PREPARATION

- Walker J**, Aparicio P, Saxena S, Hatsopoulos N, MacLean J (2026) Natural prey capture reveals flexible, feedback-driven reconfiguration of canonical motor-cortical dynamics.
- Walker J**, MacLean J, Hatsopoulos N, Saxena S (2026) Dynamics of Sensorimotor Control in a Virtual Marmoset.

CONFERENCE PRESENTATIONS

- Walker J**, MacLean J, Hatsopoulos N, Saxena S (2026) Whole body modeling of sensorimotor control during prey capture by freely moving marmosets. BRAIN initiative. BRAIN Initiative Conference. Rockville, Maryland.
- Zhang Y, **Walker J**, Hatsopoulos N, MacLean J, Saxena S (2026) Uncovering Multi-Regional Inputs Using Input-Driven Switching Recurrent Neural Networks. Cognitive Computational Neuroscience. New York City, New York.
- Wise T*, **Walker J***, Shi W, Albanese S, Johnson A, Jadi M, Nandy A, Saxena S, Chang S (2026) Developing a multimodal approach to examine cooperation and competition in Common marmosets. 2nd Annual Brain Behavior Quantification and Synchronization Workshop. Cambridge, Massachusetts.
- Walker J**, MacLean J, Hatsopoulos N, Saxena S (2026) Embodied simulation of natural behavior. *Simulated Bodies: Whole-body biomechanical models*. Janelia Research Campus. Ashburn, Virginia.
- Yongxu Z, **Walker J**, Hatsopoulos N, MacLean J, Saxena S (2026) Uncovering Multi-Regional Inputs Using Input-Driven Switching Recurrent Neural Networks. COSYNE Lisbon, Portugal.
- Samantha Johnson, **Walker J**, Hatsopoulos N (2026) Network features underlying motor learning and skill generalization in marmosets. Neural Control of Movement. Kobe, Japan
- Walker J**, MacLean J, Hatsopoulos N, Saxena S (2025) Embodied simulation of natural behavior. *Neurocybernetics at Scale*. Champalimaud Research Symposium. Champalimaud Center for the Unknown, Lisbon.

- Walker J**, MacLean J, Hatsopoulos N (2024) Building a whole -body marmoset model for deep reinforcement learning driven musculoskeletal simulation to understand sensorimotor control. *Simian Collective*. Pittsburgh.
- Walker J**, Hatsopoulos N (2023) Building a whole-body marmoset model for deep reinforcement learning driven musculoskeletal simulation to understand sensorimotor control. *Simulated Bodies: Whole body biomechanical models*. Janelia Research Campus, Ashburn, Virginia.
- Walker J**, Paul Aparicio, Jason N. Maclean, Nicholas G. Hatsopoulos. (2023) Toward understanding the dynamics of feedback control during prey capture. *Marmoset Biosciences Symposium*. Washington D.C.
- Moore D*, **Walker J***, Aparicio P, Tortolani A, MacLean J, Hatsopoulos N (2023) A sensorimotor cortical encoding model incorporating kinematic trajectories and network effects during natural forelimb movements of the marmoset. *Simian Collective*. Chicago.
- Walker J**, Moore D, MacLean J, Hatsopoulos N (2022) Understanding sensorimotor control through marmoset prey capture. *Society for Neuroscience Annual Meeting*. San Diego.
- Aparicio P, **Walker J**, Tortolani A, Hatsopoulos N (2022) 2D Spatial structure in correlations across sensorimotor cortex of freely behaving marmosets. *Society for Neuroscience Annual Meeting*. San Diego.
- Moore D*, **Walker J***, Hatsopoulos N, MacLean J (2022) Developing a network encoding model for natural behavior in the motor cortex of the unconstrained marmoset. *ARIADNE*, Santorini, Greece.
- Walker J**, MacLean J, Hatsopoulos N (2021) Characterizing neuromechanical strategies used by marmosets during foraging and prey capture. *Society for Neuroscience*. Online.
- Walker J**, Bachschmid Romano L, Moore D, Aparicio P, Sundiang M, Brunel N, Hatsopoulos N, MacLean J (2021) The importance of spatially dependent neuronal interactions in modeling cortical activity during unconstrained ethological motor behaviors in the marmoset. NIH Brain Initiative Investigators meeting. Online
- Walker J**, Moore D, Sundiang M, MacLean J, Hatsopoulos N (2019) Chronic wireless neural population recordings in freely behaving marmosets. *Society for Neuroscience Annual Meeting*. Chicago.
- Sundiang M, **Walker J**, Moore D, MacLean J, Hatsopoulos N (2019) Developing a motor task platform for freely-behaving common marmosets. *Society for Neuroscience Annual Meeting*. Chicago.
- Moore D, **Walker J**, MacLean M, Hatsopoulos N (2019) Comparing pose estimation performance using DeepLabCut and marker-based tracking. *Society for Neuroscience Annual Meeting*. Chicago.
- Walker J**, Pirschel F, MacLean J, Hatsopoulos N (2018) Encoding spaces across the behavioral repertoire of the common marmoset. *ARIADNE*, Santorini, Greece.
- Walker J**, Pirschel F, MacLean J, Hatsopoulos N (2018) Sensorimotor cortical population responses across natural behaviors. *Society for Neuroscience Annual Meeting*. San Diego.
- Walker J**, Hatsopoulos N (2015) Design of an experimental platform for voluntary behavioral training and neurophysiological study of the common marmoset. *Society for Neuroscience Annual Meeting*. Chicago.
- Walker J**, Leung S, Aaron G (2009) The Propagation of epileptiform events across the corpus callosum: results from an in vitro preparation. *American Epilepsy Society*.
- Walker J** (2007) Does inhibition increase spike time precision? *Howard Hughes Summer Research Program*.
- Walker J** (2006) The Interface of science and Buddhism: will it alter a tradition? *Wesleyan Center for East Asian Studies*.

INVITED TALKS

Whole body modeling of sensorimotor control during prey capture by freely moving marmosets. BRAIN initiative Scholar Spotlight. 2026 BRAIN Initiative Conference.	August 2026
Whole body modeling of sensorimotor control during prey capture by freely moving marmosets. Simulated Bodies: Whole-body biomechanical models. Janelia Research Campus.	April 2026
Embodied simulation of natural behavior. Neurocybernetics at Scale. Champalimaud Research Symposium.	October 2025
Inaugural Chicago Symposium on Computational Neuroscience	June 2022
IBM, Health Care Life Science Research	October 2021

HONORS AND AWARDS

Scholar Spotlight, 2026 BRAIN Initiative Conference	2026
Travel Award, Neurocybernetics at Scale at the Champalimaud Centre for the Unknown	2025
Travel Award, 10 Questions in Non-Human Primate Neuroscience at Carnegie Melon	2025
NSF Integrative Training in Motor Control and Movement (moto-IGERT)	2014-2015
Howard Hughes Summer Fellowship	Summer 2007
Mind and Life Summer Research Institute Research Fellow	Summers 2007, 2008
Wesleyan Center for the Humanities Fellowship	Fall 2007
Edward W. Snowdon Fund Wesleyan	2006
Freeman Family Grant for the Study of East Asian Diaspora	Summer 2006

SERVICE

Pathways to Science (YPRI) , mentor, Yale University	Summer 2026
Served as mentor to New Haven high school student doing summer internship in science lab at Yale.	
Initiative for Maximizing Student Development (IMSD) , peer mentor, University of Chicago	2015 – 2016
Participated in monthly discussion group around maximizing career success among graduate students from underrepresented backgrounds.	
Synapse to Self Lecture Series , <i>Co-Organizer</i> , Wesleyan University	2006 – 2007
Co-authored grant proposal as an undergraduate, with two faculty members, for an interdisciplinary lecture series about the mind and brain.	
Mind Matters: Wesleyan Journal of Psychology , <i>Senior Editor</i> , Wesleyan University	2005 – 2008
Published three annual journals of original undergraduate research in psychology and neuroscience. Organized all aspects of editorial process. Collaborated with other members of editorial board to coordinate the publication process. Maintained website. Recruited and interviewed student reviewers. Arranged and led meetings with student reviewers.	

TEACHING EXPERIENCE

Yale University

Mentor, high school student from New Haven	Summer 2026
Mentor, three graduate students from Yale University	2025-present

University of Chicago

Mentor, three graduate students from the University of Chicago	2017-2024
Mentor, four undergraduates from University of Chicago and one from Tulane	2014-2024
Teaching assistant, Brains! workshop with local primary school	Dec 2015
Participant, Course Design and College Teaching, University of Chicago Center for Teaching	Spring 2014
Teaching assistant, Computational approaches to cognitive neuroscience	Spring 2014
Teaching assistant, Mathematical methods in the biological sciences	Fall 2013

Wesleyan University

Teaching Assistant, Behavioral Neurobiology	Fall 2008
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